how it could be

Wave-shaped orbital coordinates, projection,
and rational phase addressing in UNIQUANT

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6 November 2026

Mathematical working paper, version 1.0.0; not peer reviewed

Abstract

We formalize a proposed wave-shaped path around a central reference body and connect it to rational UNIQUANT time addresses. The curve has a circular top-view projection and a sinusoidal vertical coordinate. We prove that one vertical oscillation per revolution produces a planar, centered ellipse, whereas two or more produce a nonplanar curve. Projection identities explain how apparent flattening and figure-eight crossings can arise without spatial discontinuities. We distinguish quarter-cycle address changes from physical jumps, derive curvature, torsion, exact rounding bounds, and a smooth alternative to piecewise-linear clock interpolation. A force-residual calculation determines the additional acceleration needed relative to a central inverse-square field. Reproducible synthetic checks and a chronological fitting protocol specify how the construction could be tested. No planetary fit, new force, quantum spacetime, or revision of astronomical distances is established.

Keywords: orbital geometry; sinusoidal embedding; inverse projection; rational time; phase quantization; central forces; falsifiable modeling.

1 Question, scope, and mathematical layers

The motivating question is whether an apparently circular orbital motion could be represented as a spatial wave whose appearance changes with viewing direction, with transitions marked every 90° in a top view. An earlier UNIQUANT manuscript supplies a rational time unit, annual anchors, and an explicit rounding remainder [1]. The present paper connects those ingredients to a precisely specified curve.

Four layers must be distinguished. A *geometric path* specifies possible positions. A *time parametrization* specifies how quickly that path is traversed. An *observation operator* specifies what a camera or instrument records. A *dynamical law* specifies which acceleration is physically produced. Agreement within one layer does not establish agreement within the others.

The central body is called the Sun in the diagrams, but the construction initially uses arbitrary length units. We analyze one representative body rather than assigning fitted trajectories to the planets. The numerical amplitude $A/R = 0.48$ is chosen for visibility and is not an estimate for Earth. The term *Quant* denotes the defined UNIQUANT raster unit; it carries no assertion of physical quantum discreteness. All numerical examples in this paper are synthetic.

2 A closed spatial wave around the origin

2.1 Definition and phase conventions

Let $R > 0$, $A \in \mathbb{R}$, $m \in \{1, 2, \ldots\}$, and $\delta \in \mathbb{R}$. Define

$$\mathbf{r}(\theta) = \big(R\cos\theta,\ R\sin\theta,\ A\sin(m\theta + \delta)\big), \qquad 0 \le \theta < 2\pi. \tag{1}$$

The horizontal radius is R, the vertical amplitude is $|A|$, m counts vertical oscillations per revolution, and δ is a vertical phase offset. The origin is the central reference point. For a continuous revolution count u,

$$k = \lfloor u \rfloor, \qquad \varphi = u - k \in [0, 1), \qquad \theta = 2\pi\varphi. \tag{2}$$

The integer k retains cycle order. Using $\theta = 2\pi u$ gives the same position while retaining an unwrapped phase for derivatives.

The curve is closed and smooth. Its speed with respect to phase satisfies

$$\big\|\mathbf{r}'(\theta)\big\|^2 = R^2 + A^2 m^2 \cos^2(m\theta + \delta) \ge R^2 > 0. \tag{3}$$

It is therefore regular everywhere. Moreover, (x, y) uniquely determines θ modulo 2π, so distinct phases within one revolution cannot meet at the same spatial point. Continuous reduction of A to zero preserves this property and deforms the curve into a circle. Apparent crossings in an image do not make this spatial curve a knot.

2.2 Which waves are genuinely nonplanar?

Proposition 2.1 (Planarity criterion)**.** *For the assumptions above, the path lies in a plane if and only if $A = 0$ or $m = 1$.*

Proof. If $A = 0$, the path lies in $z = 0$. If $m = 1$, the addition formula gives

$$z = \frac{A}{R}\big(\sin\delta\, x + \cos\delta\, y\big). \tag{4}$$

Conversely, suppose a plane with nonzero normal satisfies

$$\alpha R\cos\theta + \beta R\sin\theta + \gamma A\sin(m\theta + \delta) = d$$

for every θ. Averaging gives $d = 0$. For $m \ge 2$, orthogonality of Fourier modes forces $\alpha = \beta = 0$ and $\gamma A = 0$. When $A \ne 0$, this contradicts a nonzero normal. □

For $m = 1$, set $L = \sqrt{R^2 + A^2}$ and choose orthonormal in-plane vectors

$$\mathbf{e}_1 = (\cos\delta, -\sin\delta, 0), \qquad \mathbf{e}_2 = \frac{(R\sin\delta, R\cos\delta, A)}{L}.$$

Then

$$\mathbf{r}(\theta) = R\mathbf{e}_1\cos(\theta + \delta) + L\mathbf{e}_2\sin(\theta + \delta). \tag{5}$$

Thus the path is an ellipse with semiaxes R and L, centered at the origin. Its plane inclination to XY obeys $\tan i = |A|/R$. Except in the circular case $A = 0$, the Sun at the origin is its center rather than a focus. This ellipse is consequently not automatically a Kepler orbit.

For $m = 2$ and $\delta = 0$, an equivalent implicit description is

$$x^2 + y^2 = R^2, \qquad z = \frac{2A}{R^2} xy. \tag{6}$$

For $A \neq 0$, the path is the intersection of a circular cylinder and a hyperbolic paraboloid; at $A = 0$, the second surface is the plane $z = 0$. This gives a surface-based construction of the same curve without introducing cones.

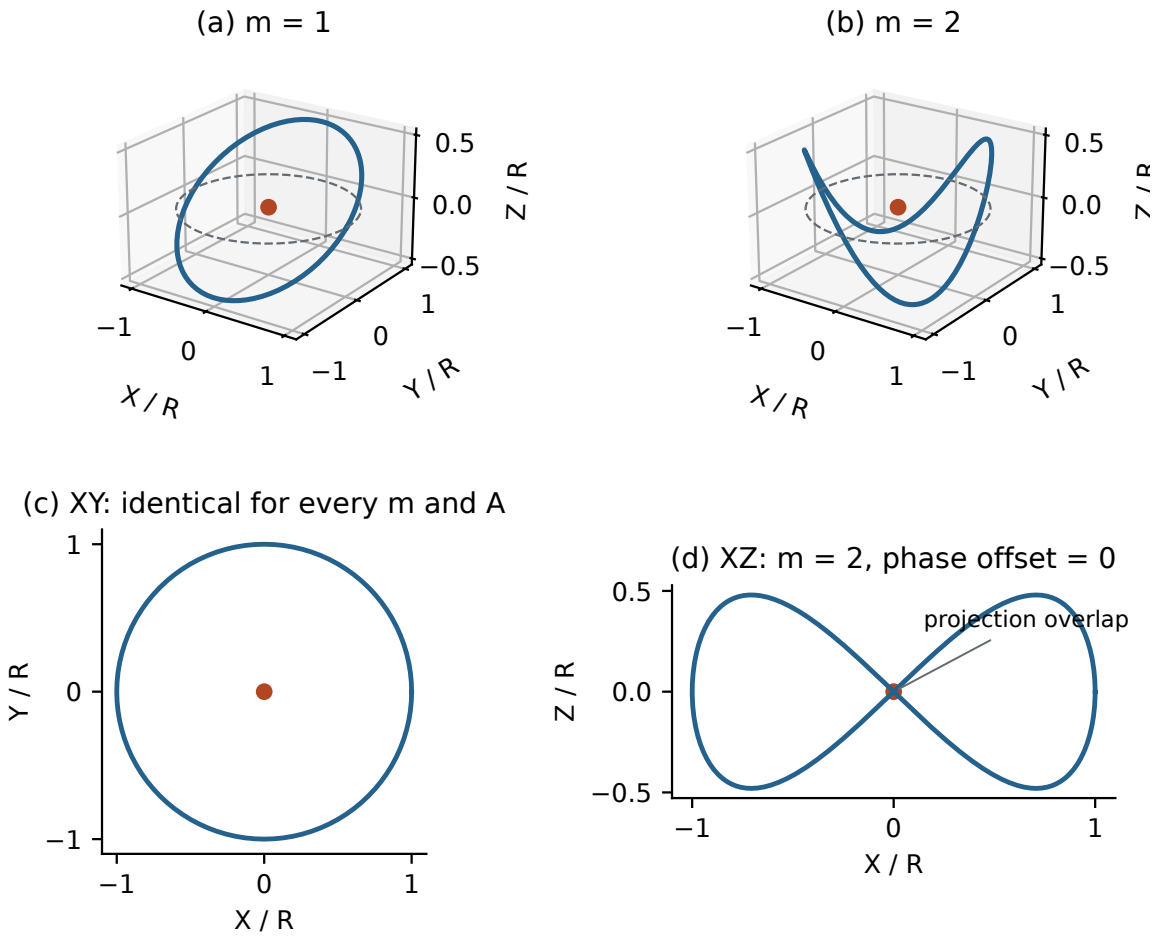


Figure 1: Synthetic geometry with $R = 1$, $A = 0.48$, and $\delta = 0$. The origin is the central reference body. (a) One vertical oscillation gives a planar ellipse. (b) Two give a nonplanar closed curve. Dashed traces are XY projections. (c) Both curves have the same circular XY projection. (d) The XZ figure-eight crossing combines two different spatial points. Lengths are normalized by R.

3 Projection, hidden depth, and reconstruction

3.1 Orthogonal views of the same path

The orthogonal projection operators are

$$P_{XY} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad P_{XZ} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{7}$$

For every A, m, δ,

$$P_{XY}\mathbf{r} = (R\cos\theta, R\sin\theta), \qquad x^2 + y^2 = R^2. \tag{8}$$

Consequently an XY image alone cannot identify the vertical amplitude, harmonic number, or vertical phase in this family. More generally, replacing z by any periodic function $f(\theta)$ leaves this projection unchanged. This is an explicit nonidentifiability result for the chosen observation operator.

For $m = 1$ and $\delta = 0$, the XZ projection is an ellipse:

$$\frac{x^2}{R^2} + \frac{z^2}{A^2} = 1 \qquad (A \neq 0).$$

For $m = 2$ and $\delta = 0$, elimination of y from (6) yields

$$z^2 = \frac{4A^2x^2}{R^2}\left(1 - \frac{x^2}{R^2}\right). \tag{9}$$

Its apparent crossing at $(x, z) = (0, 0)$ comes from the phases $\pi/2$ and $3\pi/2$. The corresponding spatial points are $(0, R, 0)$ and $(0, -R, 0)$, separated by $2R$. Projection overlap is therefore not a collision or a physical jump.

3.2 What additional information restores depth?

Stacking the two calibrated operators gives

$$H = \begin{pmatrix} P_{XY} \\ P_{XZ} \end{pmatrix}, \qquad H^{\mathsf{T}}H = \mathrm{diag}(2, 1, 1). \tag{10}$$

The combined operator has rank three. Two views therefore reconstruct XYZ when they identify the same point at the same time with known scale, orientation, and origin. Without synchronization or correspondence, the rank calculation alone is insufficient.

A real astronomical observation is not generally an orthographic image centered on the Sun. In a simplified direction-only model, let $\mathbf{d} = \mathbf{r}_{\mathrm{target}}(t_e) - \mathbf{r}_{\mathrm{obs}}(t_r)$ and

$$\mathbf{n} = \frac{\mathbf{d}}{\|\mathbf{d}\|}, \qquad t_r - t_e = \frac{\|\mathbf{d}\|}{c}. \tag{11}$$

At a fixed emission state, the direction Jacobian is

$$\frac{\partial \mathbf{n}}{\partial \mathbf{d}} = \frac{I - \mathbf{n}\mathbf{n}^{\mathsf{T}}}{\|\mathbf{d}\|}. \tag{12}$$

It has rank two and annihilates radial displacement. Independent ranging, range-rate measurements, changing observing geometry, or a specified dynamical model can constrain that missing component. Equation (11) omits several corrections required for precision astrometry; the Horizons documentation distinguishes geometric, light-time-corrected, and aberration-corrected outputs [3, 4]. Projection ambiguity by itself does not show that astronomical distances are wrong.

4 Quarter-cycle transitions and differential geometry

4.1 A changing address is not a changing trajectory

Define the zero-based quarter address

$$j(\varphi) = \lfloor 4\varphi \rfloor \in \{0, 1, 2, 3\}. \tag{13}$$

It changes at $\theta = 0, \pi/2, \pi, 3\pi/2$ modulo a revolution. With a smooth time law, the curve and its derivatives remain continuous there. A discontinuous integer label introduces no force or impulse.

For $\delta = 0$, the heights at those four boundaries are

$$m = 1 : \ (0, A, 0, -A), \qquad m = 2 : \ (0, 0, 0, 0). \tag{14}$$

Vertical extrema satisfy $m\theta + \delta = \pi/2 + \ell\pi$. Thus the $m = 2$ extrema occur at 45°, 135°, 225°, and 315°. The requested 90° partition is not an amplitude threshold or a derived transition law.

A proposed *physical* jump would require an additional hybrid model: a transition condition, a state reset, and a conservation or impulse rule. A velocity jump $\Delta\mathbf{v}$ for mass M requires an impulse $M\Delta\mathbf{v}$. None is implied by (13).

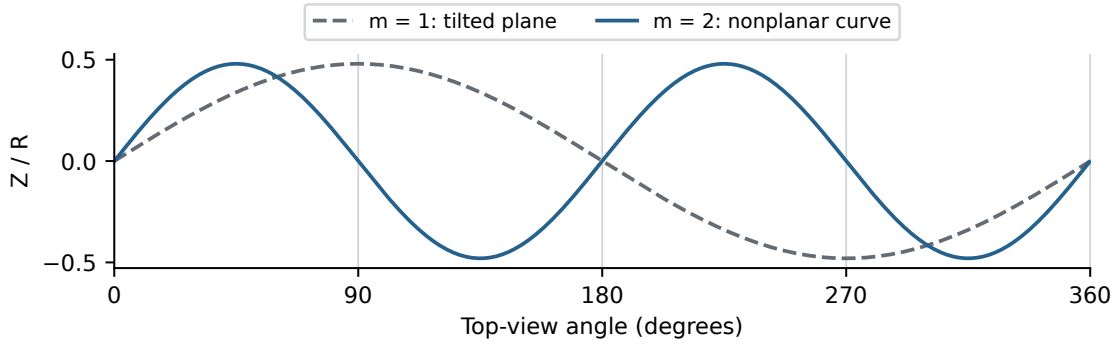


Figure 2: Height versus top-view angle for the two constructions. Vertical guides mark the requested quarter boundaries. The $m = 2$ curve crosses the reference plane at every quarter and reaches its extreme heights halfway between those boundaries. These are continuous functions.

4.2 Curvature and torsion as coordinate-independent diagnostics

Write $z = A\sin(m\theta + \delta)$ and use primes for derivatives with respect to θ. In the cylindrical orthonormal basis,

$$\mathbf{r}' = R\mathbf{e}_\phi + z'\mathbf{e}_z, \qquad \mathbf{r}'' = -R\mathbf{e}_r + z''\mathbf{e}_z, \qquad \mathbf{r}''' = -R\mathbf{e}_\phi + z'''\mathbf{e}_z.$$

Hence

$$\left\|\mathbf{r}' \times \mathbf{r}''\right\|^2 = R^2(R^2 + z'^2 + z''^2), \qquad (\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}''' = R^2(z' + z''').$$

The curvature κ and torsion $\mathcal{T}$ are therefore

$$\kappa(\theta) = \frac{R\sqrt{R^2 + z'^2 + z''^2}}{(R^2 + z'^2)^{3/2}}, \tag{15}$$

$$\mathcal{T}(\theta) = \frac{z' + z'''}{R^2 + z'^2 + z''^2} = \frac{Am(1 - m^2)\cos(m\theta + \delta)}{R^2 + A^2m^2\cos^2(m\theta + \delta) + A^2m^4\sin^2(m\theta + \delta)}. \tag{16}$$

Torsion vanishes identically for $m = 1$. For $m = 2$ and $A \neq 0$, it does not; no fixed spatial rotation can make the entire curve planar. Isolated zeros of torsion do not establish global planarity. Both quantities have units of inverse length and describe geometry rather than a force law.

5 Connecting the phase to UNIQUANT time

5.1 Rational unit, address, and exact remainder

We retain the earlier manuscript's definitions [1]:

$$q = \frac{86400}{768 \cdot 66 \cdot 23}\ \mathrm{s} = \frac{75}{1012}\ \mathrm{s} \approx 0.07411067194\ \mathrm{s}. \tag{17}$$

For a dimensionless model time τ in units of q, define

$$\mathsf{A}(\tau) = \left\lfloor \tau + \frac{1}{2} \right\rfloor, \qquad \rho(\tau) = \tau - \mathsf{A}(\tau), \qquad -\frac{1}{2} \le \rho < \frac{1}{2}. \tag{18}$$

The symbol A denotes an integer address and is distinct from the geometric amplitude A. Retaining both A and ρ recovers τ exactly. The displayed time is $q\mathsf{A}$; the complete model time is $q(\mathsf{A} + \rho)$.

Proposition 5.1 (Address and interval bounds)**.** *For any model times τ_a, τ_b,*

$$|q\mathsf{A}(\tau) - q\tau| \le q/2, \tag{19}$$

$$\left|q\big(\mathsf{A}(\tau_b) - \mathsf{A}(\tau_a)\big) - q(\tau_b - \tau_a)\right| < q. \tag{20}$$

Proof. The first bound follows from the remainder interval in (18). The signed interval error is $q(\rho_a - \rho_b)$; two members of $[-1/2, 1/2)$ differ by less than one. □

These are representation bounds, not bounds on astronomical model error. If a model trajectory has speed bounded by V over the relevant interval, its position change caused by discarding ρ is at most $Vq/2$. Preserving the remainder avoids that particular loss without correcting a wrong trajectory.

5.2 Annual anchors and a continuous phase convention

The specified annual anchor is

$$\tau(n) = Bn + \frac{an - bn^2}{D}, \quad (B, a, b, D) = (425806080, 703993, 11, 529). \tag{21}$$

Equivalently, $P(n) = DBn + an - bn^2$ is an integer and $\tau(n) = P(n)/D$. The coefficients are inherited model definitions, not re-estimated astronomical quantities.

To reproduce the interactive construction, identify one model year with one geometric revolution and interpolate linearly between its anchors:

$$\tau_{\mathrm{lin}}(u) = \frac{P(k) + \varphi\,[P(k+1) - P(k)]}{D}, \quad u = k + \varphi. \tag{22}$$

This identification and interpolation are additional conventions. They do not follow from the rounding theorem. The corresponding cycle duration is

$$T_k = q\left[B + \frac{a - b(2k+1)}{D}\right]. \tag{23}$$

Only a declared interval on which the relevant time map is increasing may be used. No indefinite physical extrapolation of the quadratic annual model is justified.

The earlier epoch label is 13 July 2029 at 00:00 MESZ, equivalently 12 July 2029 at 22:00 UTC [1]. Here we use relative model time. Applying the construction to ephemerides requires an explicit conversion to a declared working time scale; a civil epoch label alone is not a TDB realization.

5.3 A smooth alternative and its exact discrepancy

Equation (22) is continuous but has different slopes on adjacent model years. If treated as a physical clock-to-orbit law, those changes would produce velocity jumps. To avoid inserting such artifacts, one can instead use the smooth extension

$$\tau_{\mathrm{sm}}(u) = Bu + \frac{au - bu^2}{D}. \tag{24}$$

Both extensions agree at integer anchors, while direct expansion gives

$$\tau_{\mathrm{sm}}(u) - \tau_{\mathrm{lin}}(u) = \frac{b\varphi(1-\varphi)}{D}, \qquad 0 \le q(\tau_{\mathrm{sm}} - \tau_{\mathrm{lin}}) \le \frac{qb}{4D} \approx 0.385263 \text{ ms}. \tag{25}$$

This is a difference between two chosen interpolations, not an observational error estimate.

For $t = q\tau_{\mathrm{sm}}(u)$ and unwrapped angle $\theta = 2\pi u$, define

$$g(u) = B + \frac{a - 2bu}{D} > 0.$$

The chain rule gives

$$\dot\theta = \frac{2\pi}{qg(u)}, \qquad \ddot\theta = \frac{4\pi b}{Dq^2 g(u)^3}, \qquad \ddot{\mathbf{r}} = \mathbf{r}''\dot\theta^2 + \mathbf{r}'\ddot\theta. \tag{26}$$

Thus a time reparametrization changes the acceleration assigned to a fixed geometric curve. A formula derived for constant angular speed must not be silently reused with this nonuniform clock.

5.4 The 13/23 partition and quarter boundaries

The combined phase boundaries $j/13$ and $\ell/23$ are represented exactly by

$$\mathcal{B} = \mathrm{sort}(\{23j : 0 \le j \le 13\} \cup \{13\ell : 0 \le \ell \le 23\})\,/299. \tag{27}$$

There are 36 distinct boundaries and 35 intervals. Coprimality implies that only the endpoints are shared. Sorting the integer numerators verifies the interval-width multiset

$$2 \times \left\{\frac{1}{299}, \ldots, \frac{12}{299}\right\}, \qquad 11 \times \left\{\frac{13}{299}\right\}. \tag{28}$$

The three internal quarter boundaries do not belong to this set: $j/4 = n/299$ would require $299j = 4n$, impossible for $j = 1, 2, 3$. Quarter labels and 13/23 class labels must therefore be stored separately. A chart subdivision adds no independent dynamical degree of freedom.

At integer annual anchors the centered remainder has least period 529. Indeed, a residue period h requires $P(n+h) - P(n)$ to be divisible by 529 for every integer n. Comparing consecutive n gives $529 \mid 22h$; since $\gcd(22, 529) = 1$, $529 \mid h$, and $h = 529$ works. This arithmetic periodicity is not a 529-year physical orbit or energy cycle.

6 Which dynamics could generate the geometric curve?

6.1 Uniform-phase kinematics and a force residual

For this section only, assume $\theta = \omega t$, with constant $\omega > 0$ and $\delta = 0$. Writing $(x, y, z) = \mathbf{r}(t)$,

$$\dot{\mathbf{r}} = \omega(-R\sin\theta,\ R\cos\theta,\ mA\cos m\theta), \tag{29}$$

$$\ddot{\mathbf{r}} = -\omega^2(x, y, m^2 z). \tag{30}$$

For a Newtonian central reference field with parameter μ,

$$\mathbf{a}_N = -\frac{\mu}{d^3}\mathbf{r}, \qquad d = \|\mathbf{r}\| = \sqrt{R^2 + z^2}, \qquad \beta = \frac{\mu}{d^3}. \tag{31}$$

The additional acceleration needed to realize the prescribed path is exactly

$$\boxed{\mathbf{a}_{\text{add}} = \big((\beta - \omega^2)x,\ (\beta - \omega^2)y,\ (\beta - m^2\omega^2)z\big).} \tag{32}$$

If a central coefficient were set to $\beta = \omega^2$, the remaining vertical term would be $-(m^2 - 1)\omega^2 z$, equal to $-3\omega^2 z$ for $m = 2$. For the actual inverse-square expression, β generally varies and all terms in (32) must be retained.

6.2 Angular momentum excludes a sole central-force explanation

The specific angular momentum $\mathbf{h} = \mathbf{r} \times \dot{\mathbf{r}}$ obeys

$$\dot{\mathbf{h}} = \mathbf{r} \times \ddot{\mathbf{r}} = (m^2 - 1)\omega^2 z(-y, x, 0). \tag{33}$$

Any force directed to one fixed center has zero torque about that center. Equation (33) therefore rules out a sole central-force interpretation for the nonplanar $m = 2$ path with $A \neq 0$. More generally, a nonzero conserved angular momentum confines central-force motion to a fixed plane; changing the rate of traversal cannot remove that geometric obstruction.

For $m = 1$, (30) instead gives $\ddot{\mathbf{r}} = -\omega^2\mathbf{r}$. This is a harmonic central acceleration. It equals an inverse-square acceleration throughout the orbit only if d is constant. Since $d^2 = R^2 + A^2 \sin^2\theta$, that requires $A = 0$ in this family; the circular case permits $\omega^2 = \mu/R^3$. Planarity alone does not make the centered ellipse a solar Kepler solution.

These statements concern an ideal fixed-center field. Planetary perturbations and consistently included moving-frame terms can produce nonplanarity. They must be modeled before attributing any measured residual to an additional force. A general noninertial frame can change apparent paths; its inertial terms cannot be omitted from a dynamical comparison.

6.3 A consistent conservative realization

One mathematical dynamics that generates (30) is the anisotropic harmonic potential per unit mass

$$\Phi_{\text{osc}}(x, y, z) = \frac{1}{2}\omega^2(x^2 + y^2 + m^2 z^2). \tag{34}$$

Its negative gradient is the required acceleration, and the conserved specific energy is

$$E_{\text{osc}} = \frac{1}{2}\|\dot{\mathbf{r}}\|^2 + \Phi_{\text{osc}} = \omega^2 R^2 + \frac{1}{2}m^2\omega^2 A^2. \tag{35}$$

The horizontal and vertical angular frequencies are ω and $m\omega$; their ordinary frequencies are $\omega/(2\pi)$ and $m\omega/(2\pi)$. Their periods are $2\pi/\omega$ and $2\pi/(m\omega)$. An observation window and a sampling interval are separate quantities: they restrict frequency estimation and may create aliasing but do not define a unique physical frequency by themselves. This construction is a commensurate classical oscillator. It supplies neither a source for this potential in the Solar System nor evidence for two physical energy waves.

Relative to the Newtonian potential, define

$$E_N = \frac{1}{2}\omega^2(R^2 + m^2 A^2 \cos^2 m\theta) - \frac{\mu}{d}.$$

Differentiation yields the explicit work balance

$$\dot{E}_N = \dot{\mathbf{r}} \cdot \mathbf{a}_{\text{add}} = m\omega A^2 \sin m\theta \cos m\theta \left(\frac{\mu}{d^3} - m^2\omega^2 \right). \tag{36}$$

Changes in this reference energy represent work by the additional acceleration. They cannot by themselves be interpreted as a change of orbital branch or as a mechanism for climate variation.

7 Mathematical approaches to fitting and falsification

7.1 Separate orientation, deformation, and clock

A useful extension is a finite harmonic height model,

$$z(\theta) = c + \sum_{\ell=1}^{M} \left[a_\ell \cos(\ell\theta) + b_\ell \sin(\ell\theta) \right]. \tag{37}$$

For a circular XY trace, the $\ell = 1$ terms describe a tilted plane, while higher harmonics can describe nonplanar deformation. In an inertial coordinate fit, orientation must be estimated before interpreting a first-harmonic vertical signal as new physics. A plane through the declared origin can be fitted by

$$\min_{\|\mathbf{n}\|=1} \sum_i w_i (\mathbf{n} \cdot \mathbf{r}_i)^2. \tag{38}$$

The minimizing normal is an eigenvector for the smallest eigenvalue of $\sum_i w_i \mathbf{r}_i \mathbf{r}_i^{\mathsf{T}}$. With correlated or anisotropic errors, a full covariance-aware fit is needed rather than these scalar weights.

For fixed known phases, (37) is linear in its coefficients. Generalized least squares gives

$$\widehat{\boldsymbol{\alpha}} = (X^{\mathsf{T}}\Sigma^{-1}X)^{-1}X^{\mathsf{T}}\Sigma^{-1}\mathbf{z}, \tag{39}$$

provided Σ is positive definite and the design matrix has full column rank. Harmonic amplitude and phase are

$$A_\ell = \sqrt{a_\ell^2 + b_\ell^2}, \qquad \delta_\ell = \operatorname{atan2}(a_\ell, b_\ell).$$

If phase, frequency, reference frame, or time mapping is also unknown, fitting is generally nonlinear and the joint identifiability problem must be checked. A well-fitted curve is an interpolation model until it makes independently tested predictions.

7.2 A concrete astronomical test protocol

An empirical test should use a specified baseline and observation model:

1. Declare the target body, coordinate origin, inertial frame, units, working time scale, and interval. Record data versions and uncertainty models.
2. First compare geometric positions and velocities with a gravitational ephemeris under matching conventions. A concrete diagnostic query could use Earth (399), center `500@10`, `EPHEM_TYPE=VECTORS`, `REF_SYSTEM=ICRF`, `REF_PLANE=FRAME`, `TIME_TYPE=TDB`, and `VEC_CORR=NONE`, with explicitly chosen dates and units [4]. These are Sun-relative states in ICRF-oriented axes, not states with an inertial origin. Dynamical comparisons must include the acceleration of that origin through indirect terms, or instead use barycentric states. This is a proposed query, not one executed for this paper.
3. Fit orientation and any clock parameters on an early training block. Select harmonic order and regularization on a later validation block. Keep the final chronological block untouched until evaluation.

4. Compare equal-information models: the baseline; a coordinate-only UNIQUANT re-expression; and a baseline with explicitly parameterized additional dynamics. For independent vector residuals $\mathbf{e}_i$, report position and velocity RMS errors and the covariance-weighted sum $\sum_i \mathbf{e}_i^{\mathsf{T}} \Sigma_i^{-1} \mathbf{e}_i$. For correlated observations, use the full stacked covariance instead of independent blocks.
5. Evaluate open forecasts without future resets. Where available, also predict independent angular positions, ranges, and range rates through the corresponding observation operators.
6. Examine the expected signatures: higher-harmonic out-of-plane residuals, vertical velocity amplitude $m\omega|A|$, vertical acceleration amplitude $m^2\omega^2|A|$, and the torque in (33). Test whether conventional perturbations, frame terms, or calibration errors explain them.

The published DE440/DE441 construction fits numerically integrated trajectories to terrestrial and spacecraft observations [2]. Fitting its exported coordinates evaluates approximation to that ephemeris. Evidence for a different physical model additionally requires robust performance on independent observations, with instrumental and model uncertainties included. No such comparison is performed here.

8 Reproducible synthetic checks

The supplement contains a Python script that regenerates all figures and the numerical table. Lengths are normalized by $R = 1$; $A = 0.48$ and $\delta = 0$. Planarity is checked by singular-value decomposition of 1,440 equally spaced positions. The best-plane RMS is approximately zero for $m = 1$ and $A/\sqrt{2}$ for this $m = 2$ example.

Exact rational arithmetic checks all 2,161 phase settings of the demonstration, at half-degree spacing across three model cycles. It checks the address bounds, negative half-integer tie convention, annual remainder periodicity, and the 35 phase cells. Floating-point evaluations separately verify the oscillator energy and torque formulas. Algebraic proofs, not numerical tolerances, establish the general results.

For a fitting example, generate three cycles of

$$z_i = 0.48 \sin(2\theta_i) + \varepsilon_i, \qquad \varepsilon_i \sim N(0, 0.015^2), \tag{40}$$

with 720 points per cycle and fixed random seed 23. The first two cycles train two least-squares models: a constant plus first harmonic, and the same model with a second harmonic. The final cycle is held out. The resulting error reduction is expected because the second harmonic was explicitly inserted into the synthetic generator. It demonstrates the fitting procedure, not an astronomical discovery.

Table 1: Deterministic reproduction results. Geometric RMS errors and fitted amplitude use units of R; time quantities have explicit units. RMSE values compare predictions with noisy held-out synthetic heights.

Quantity	Value
Best-plane RMS, $m = 1$	5.98e-17
Best-plane RMS, $m = 2$	0.339411
Largest sampled address error (s)	0.037047553
Universal address bound $q/2$ (s)	0.037055336
Largest linear/smooth clock difference (ms)	0.385263
Held-out RMSE, plane-only model	0.339776
Held-out RMSE, model with $m = 2$	0.014891
Fitted $m = 2$ amplitude (true value 0.48)	0.479644

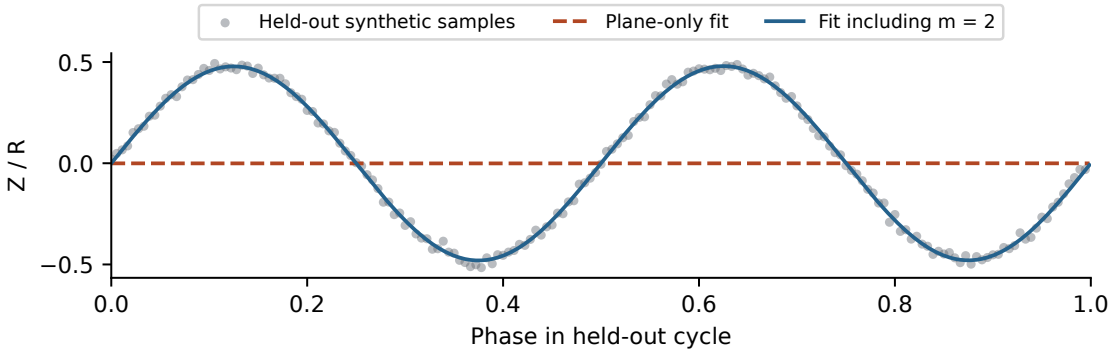


Figure 3: Held-out synthetic cycle. The second-harmonic model recovers the deliberately inserted vertical signal; a plane-only model cannot represent it. Samples are thinned for display only. No astronomical observations enter this figure.

9 Interpretation and next admissible claims

The construction establishes a family of regular closed curves, its planarity boundary, projection ambiguities, exact rational addressing, and a dynamically consistent oscillator realization. It also supplies a precise obstruction: a nonplanar member cannot be produced by a sole fixed-center central force. These are mathematical statements about the declared model.

It remains unknown whether a fitted higher-harmonic term would improve an appropriate astronomical baseline. A 90° phase boundary is currently an indexing convention. A 529-periodic rounding remainder is arithmetic. Neither is an observed transition of a planet. The word *wave* in a coordinate function does not establish a propagating field, a medium, a quantization of matter, or a hidden spatial dimension.

The next defensible step is a covariance-aware fit in a declared inertial frame, followed by a chronological comparison and a check of the required acceleration against established perturbations. A physical transition hypothesis would additionally need an explicit triggering condition, state evolution, and independent observable signature. The present framework makes those missing ingredients identifiable and calculable without assuming their existence.

A Exact address calculation on the demonstration grid

For a grid with N subdivisions per revolution, let $s = Nk + f$, $0 \le f < N$. Define

$$p_s = NP(k) + f\,[P(k+1) - P(k)], \qquad d_s = DN. \tag{41}$$

The time $\tau_s = p_s/d_s$ has the exact address and remainder

$$\mathsf{A}_s = \left\lfloor \frac{2p_s + d_s}{2d_s} \right\rfloor, \qquad \rho_s = \frac{p_s - d_s\mathsf{A}_s}{d_s}. \tag{42}$$

This expression handles negative times when integer division implements mathematical floor. Truncation toward zero is not equivalent for negative numerators.

The Python reference implementation is:

```
from fractions import Fraction
B, a, b, D = 425806080, 703993, 11, 529
def P(k):
    return D*B*k + a*k - b*k*k
def state(s, N=720):
    k, f = divmod(s, N)
    p = N*P(k) + f*(P(k+1)-P(k))
    d = D*N
    A = (2*p + d) // (2*d)
    rho = Fraction(p-d*A, d)
    return k, Fraction(f, N), A, rho
```

At a cycle boundary $s = N(k+1)$, the local phase returns to zero and the cycle counter increments. Position remains continuous. Under the piecewise-linear clock, differentiability at that boundary is a separate issue discussed in Section 5.3.

Authorship, provenance, and computational availability

The UNIQUANT naming, prior coordinate conventions, and motivating wave-orbit proposal originate in Joern Christian Anders's documented project. This manuscript develops the proposal discussed with ChatGPT and the accompanying interactive demonstration. ChatGPT (OpenAI) assisted with derivations, independent formula checks, coding, synthetic illustrations, and English drafting. AI systems are not listed as scientific coauthors. The title is retained exactly as requested by the author. The manuscript is a working draft and has not undergone independent peer review.

The accompanying source archive includes `paper.tex`, `reproduce.py`, `results.json`, vector figures, and instructions. Running `python reproduce.py` regenerates the synthetic outputs; running `pdflatex paper.tex` twice builds the paper. No Solar System data files are included because none were analyzed.

References

[1] J. C. Anders. *Uniquant: rationale Zeit- und Phasenkoordinaten. Beschränkte Quantisierungsfehler, geometrische Momente und identifizierbare Verteilungsrekonstruktion*. Mathematical working paper 1.0.0, 6 November 2026. Author-supplied manuscript; local source file `UNIQUANT_Paper_V1_0_0-1.pdf`. Not peer reviewed.

[2] R. S. Park, W. M. Folkner, J. G. Williams, and D. H. Boggs. The JPL Planetary and Lunar Ephemerides DE440 and DE441. *The Astronomical Journal* 161, 105 (2021). doi:10.3847/1538-3881/abd414. Official record: `https://ssd.jpl.nasa.gov/doc/de440_de441.html`.

[3] NASA Jet Propulsion Laboratory, Solar System Dynamics. *Horizons System Manual*. `https://ssd.jpl.nasa.gov/horizons/manual.html`. Accessed 6 November 2026.

[4] NASA Jet Propulsion Laboratory, Solar System Dynamics. *Horizons API documentation*. `https://ssd-api.jpl.nasa.gov/doc/horizons.html`. Accessed 6 November 2026.